Available online at website: https://jurnal.iaii.or.id/index.php/RESTI

JURNAL RESTI

(Rekayasa Sistem dan Teknologi Informasi)

Vol. 9 No. 4 (2025) 754 - 764 e-ISSN: 2580-0760

BERT Model Fine-tuned for Scientific Document Classification and Recommendation

Muhammad Deagama Surya Antariksa[1*], Aris Sugiharto[2], Bayu Surarso[3]

[1]Master of Information Systems, Postgraduate School, Universitas Diponegoro, Semarang, Indonesia

[2]Department of Informatics, Faculty of Science and Mathematics, Universitas Diponegoro, Semarang, Indonesia

[3]Department of Mathematics, Faculty of Science and Mathematics, Universitas Diponegoro, Semarang, Indonesia

[1]deagama@students.undip.ac.id, [2]aris.sugiharto@live.undip.ac.id, [3]bayus@lecturer.undip.ac.id

Abstract

The increasing number of academic documents requires efficient and accurate classification and recommendation systems to assist in retrieving relevant information. This system is built using the "bert-base-uncased" model from Hugging Face, which has been fine-tuned to improve the classification accuracy and relevance of document recommendations. The dataset used consists of 2.000 academic documents in the field of computer science, with features including titles, abstracts, and keywords, which were combined into a single input for the model. Document similarity is measured using cosine similarity, resulting in recommendations based on semantic proximity. Unlike traditional approaches, which rely primarily on word frequency or surface-level matching, the proposed method leverages BERT's contextual embeddings to capture deeper semantic meanings and relationships between documents. This allows for more accurate classification and more context-aware recommendations. Evaluation results show that the best model configuration (learning rate 3e-5, batch size 32, optimizer AdamW) achieved 89.5% training accuracy and an F1-score of 0.8947, while testing yielded 91% accuracy and 90% F1-score. The recommendation system consistently produced Precision@k values above 92% for k between 5 and 30, with Recall@k reaching 1.0 as k increased. These results indicate that the system not only performs reliably in classifying complex academic texts but also effectively recommends contextually relevant documents. This integrated approach shows strong potential for enhancing academic document retrieval and supports the development of semantically aware information management systems.

Keywords: BERT; cosine similarity; document classification; fine-tuning; recommendation system

How to Cite: Antariksa, M. D. S., Sugiharto, A., & Surarso, B. (2025). BERT Model Fine-tuned for Scientific Document Classification and Recommendation. *Jurnal RESTI (Rekayasa Sistem Dan Teknologi Informasi)*, *9*(4), 754 - 764. https://doi.org/10.29207/resti.v9i4.6789
*Permalink/*DOI: https://doi.org/10.29207/resti.v9i4.6789

Received: June 17, 2025
Accepted: July 23, 2025
Available Online: August 13, 2025

1. Introduction

The swift progression of information and communication technology has dramatically transformed data processing and access methods. The increasing volume of textual data poses significant management challenges [1]. To address this, various Artificial Intelligence (AI) and Machine Learning (ML) techniques have been applied for data filtering, classification, and recommendation [2]. The surge in open-access publications has further emphasized the need for intelligent systems in academic information retrieval [3].

In educational settings, AI and ML offer powerful tools for organizing diverse documents, facilitating efficient classification and recommendation processes [4]. This organization enhances information retrieval, enabling quicker access to academic content [5], and supports better knowledge management through intelligent system integration [6].

Recommender systems play a vital role in helping users locate relevant materials. Accurate recommendations improve retrieval precision and reduce search time, thereby supporting research efficiency and scientific advancement [7]. Both content-based and hybrid recommendation approaches are now commonplace in digital learning platforms [8].

Advances in Natural Language Processing (NLP) have significantly driven the development of modern classification and recommendation systems [9]. NLP has evolved from rule-based approaches to deep learning techniques, especially with the emergence of Transformer-based models that exhibit exceptional capabilities in contextual understanding [10]. The Transformer is a type of neural network that primarily uses a self-attention mechanism to extract intrinsic features, allowing each word in a sequence to respond dynamically based on similarity scores. This enables the model to capture complex dependencies both across and within sequences, making it highly effective for sequential data processing [11]. One of the most influential models built on the Transformer architecture is BERT (Bidirectional Encoder Representations from Transformers), which leverages pre-training on large corpora and self-attention mechanisms to generate rich semantic representations for various NLP tasks [12].

A notable benefit of BERT is its adaptability via fine-tuning for specific tasks, such as classification or semantic search [13]. Enhanced strategies such as layer freezing, learning rate tuning, and discriminative fine-tuning have proven effective in domain-specific applications [14]. Furthermore, adapter-based fine-tuning methods offer improved computational efficiency without sacrificing performance [15].

Traditional approaches to document classification continue to provide a solid foundation in NLP research. One study used the BBC News dataset containing 2225 documents across five categories and applied a combination of Term Frequency (TF), TF-IDF, chi-square feature selection, and the Bat algorithm, tested across Random Forest, SGD, AdaBoost, and CatBoost classifiers. Random Forest yielded the best performance with 87.57% accuracy and 87.91% F1-score, demonstrating that combining statistical feature selection with metaheuristic optimization can enhance accuracy while reducing model complexity and overfitting risk in high-dimensional data [16]. Another study employed a multilabel news dataset consisting of around 60000 articles classified into 24 labels. It used problem transformation techniques such as Binary Relevance and Label Powerset with feature extraction via Count Vectorizer and TF-IDF, evaluated using Logistic Regression, Naive Bayes, and Linear SVC. The best result came from Label Powerset with TF-IDF and Linear SVC, achieving 91% accuracy and F1-score, showing its strength in capturing label correlations and leveraging TF-IDF's representational power [17]. In the domain of short-text classification, a study used Twitter data from the Los Angeles Times account, comprising 687 entries in 20 categories. After preprocessing and TF-IDF feature extraction, five models were tested. Random Forest again performed best with 92% accuracy and 93% F1-score, particularly excelling in granular social media categories such as technology and travel, where it achieved perfect F1-scores of 1.00 [18]. Despite these results, traditional methods often struggle to capture deeper semantic relationships, particularly in complex technical texts [19].

To overcome the limitations of traditional methods, deep learning has emerged as a more powerful alternative for text classification. In a hoax detection task on Indonesian Twitter, a dataset of 25325 tweets labeled as either hoax or non-hoax was used to evaluate a Convolutional Neural Network (CNN) model combined with multiple text representations, including TF-IDF bigrams, BERT embeddings, and GloVe. The best performance was achieved by combining all three representations, resulting in an accuracy of 98.57 percent, which significantly outperformed the CNN with TF-IDF baseline that only reached 94.15 percent. This highlights the advantage of integrating diverse semantic features in social media analysis [20]. Similarly, a study on hate speech and abusive content classification in Indonesian employed Graph Neural Networks (GNN) on a manually labeled dataset of 13169 tweets. GNN achieved 92.90 percent accuracy for abusive content and 89.78 percent for hate speech, outperforming Recurrent Neural Networks and demonstrating superior capability in capturing structural and contextual relationships in short texts [21]. For low-resource languages, transformer-based models such as XLM-RoBERTa, BanglaBERT, and mBERT were applied to a dataset of 5839 social media posts in Bangla, categorized into nine emergency-related types. Among the models, XLM-RoBERTa achieved the highest F1-score of 95.25 percent, showing strong effectiveness in managing contextual dependencies and handling code-mixed language scenarios, which is especially valuable for early warning systems in linguistically underserved domains [22].

Fine-tuning Transformer-based models such as BERT has proven highly effective for tasks requiring deeper semantic understanding. A study fine-tuning BERT on 10000 policy documents from nine institutions reported an F1-score of 93.25%, outperforming CNN and BiLSTM due to architectural refinements [23]. Another study comparing BERT, RoBERTa, and BART across domains like COVID-19 hoaxes and extremist content found that BERT-base achieved 99.71% accuracy, surpassing traditional models like CNN and XGBoost by up to 24% [24]. In online review classification, fine-tuned BERT achieved 70.7% accuracy and a 71.7% F1-score, outperforming SVM, and highlighting the advantage of bidirectional contextual embeddings without heavy preprocessing [25].

In content recommendation, traditional systems often utilize TF-IDF in combination with cosine similarity to measure relevance between content items. One implementation in a streaming platform used a dataset of 7787 unique entries, analyzing title and description similarity to generate recommendations. The system achieved a highest similarity score of 0.2195 for closely related content such as the "NCIS" series, indicating potential despite being limited to only two textual features. Its efficiency could be improved further by incorporating additional attributes such as duration, rating, actors, and user demographics [26]. A similar approach was applied in a mobile-based social media application, where TF-IDF and cosine similarity were used to compare user profiles and community descriptions. Testing 80 user profiles against 10 communities yielded up to 90 percent accuracy with a 10 percent error rate. This system effectively reduced information overload and improved the discovery of relevant communities by considering user interactions, multilingual content, and a three-layer modular architecture [27].

Embedding-based approaches using BERT have shown significant improvements in recommendation quality by leveraging contextual feature extraction and semantic similarity. In an anime recommendation system, BERT was used to extract contextual features from titles and genres across a dataset of 17562 entries. The system achieved high cosine similarity scores, including 0.994 for Naruto, 0.991 for Boruto, and 0.993 for Dragon Ball Super, demonstrating its effectiveness in identifying thematically related content. However, the model exhibited a tendency to prioritize large franchises with multiple series, which affected recommendation diversity [28]. In a curriculum-driven academic paper search system, BERT and cosine similarity were applied to 4112 documents spanning mathematics, engineering, and computer science. The input queries were formulated from course topics, and document matching was performed using keyword vector encoding. The system successfully retrieved relevant documents for all topics, whereas Scopus failed to return results for up to eight topics per course. Average similarity scores based on keyword embeddings reached 0.43, compared to 0.37 for title embeddings, underscoring the greater semantic effectiveness of keyword-based representations in academic search tasks [29].

Although various approaches have been successfully applied to classification and recommendation tasks, most prior studies tend to address these components separately. There remains a gap in integrated approaches that combine classification and recommendation systems based on semantic embeddings using BERT models, particularly in the context of scientific documents sourced from institutional repositories in Indonesia. To bridge this gap, an integrated system based on a fine-tuned BERT model was developed, capable of generating semantic representations and measuring document similarity using cosine similarity. This system is specifically designed for computer science documents and evaluated using both classification metrics (accuracy, F1-score) and recommendation relevance metrics (Precision@k, Recall@k), thus contributing to context-aware academic information management.

Therefore, the objective of this study is to design and evaluate a unified system that performs both document classification and recommendation using a fine-tuned BERT model. The system aims to improve classification accuracy through deeper semantic understanding and enhance recommendation relevance by leveraging semantic embeddings and cosine similarity, particularly for complex academic documents in the computer science domain.

2. Methods

The methods in this study consisted of a series of research procedures designed to develop two integrated systems, namely a document classification system and a recommendation system, both based on the BERT model optimized for specific tasks. The overall methodology followed standard practices in the field of Natural Language Processing (NLP), structured systematically from data collection to final evaluation. Each stage was executed incrementally to ensure reproducibility and methodological rigor.

The initial focus of the research was the development of a document classification system. The complete process is illustrated in Figure 1.

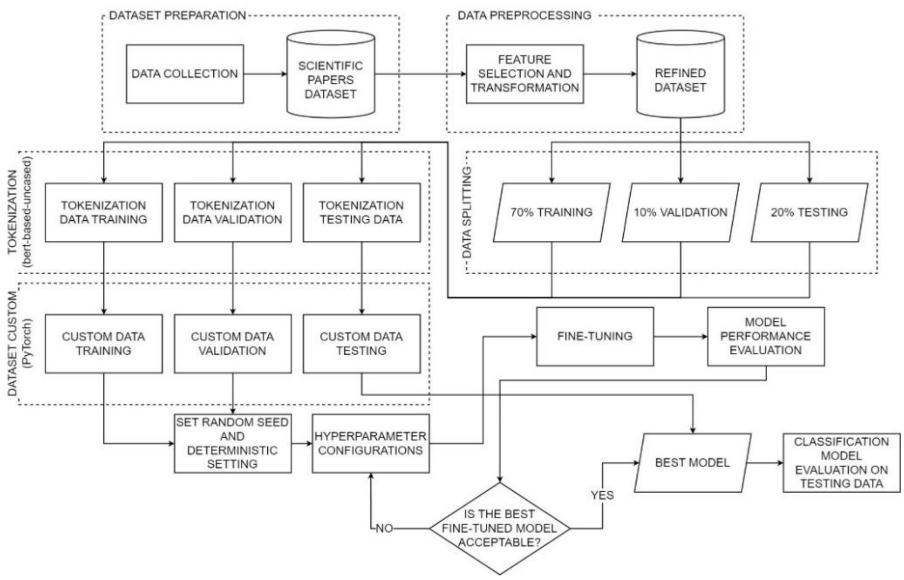


Figure 1. Research Procedure for Classification System

Figure 1 illustrates the workflow for scientific document classification using the BERT-base-uncased model. The process began with data collection from institutional repositories, consisting of titles, abstracts, and keywords. These three components were merged into a unified textual representation, followed by a series of preprocessing steps such as cleaning and normalization. Each document was then analyzed and assigned a class label based on its content, which was converted into numerical format. The resulting dataset was split into three subsets: 70% for training, 10% for validation, and 20% for testing.

The BERT-base-uncased model was fine-tuned using various hyperparameter configurations. During training, the model's performance was monitored using loss, accuracy, and F1-score on both the training and validation sets. The best model was selected not solely based on the highest accuracy or F1-score, but also by evaluating the stability of these metrics across validation runs. Additionally, loss trends were observed to ensure that performance improvements were consistent and the model did not exhibit signs of overfitting. This approach ensured that the selected model had a good fit and generalization ability. Final evaluation on the test set was conducted using accuracy, precision, recall, F1-score, and a confusion matrix to comprehensively assess the model's classification performance.

After completing the classification stage, the research continued with the development of a document recommendation system. The series of processes involved in this stage is presented in Figure 2.

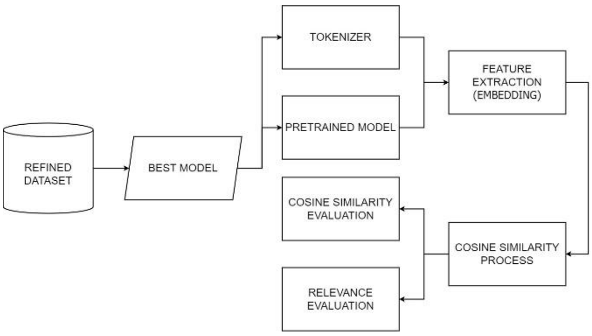


Figure 2. Research Procedure for Recommendation System

Figure 2 explains the development of a document recommendation system based on semantic similarity. The system was built using a refined dataset, which was the result of merging and preprocessing the textual elements of each document. This data was then reprocessed using the tokenizer and pretrained architecture of the fine-tuned BERT-base-uncased model from the classification phase. Tokenization was performed to match the input format expected by the model, followed by feature extraction using the [CLS] token embedding to represent the semantic meaning of each document.

The extracted embeddings were used to calculate cosine similarity between documents in order to assess semantic closeness. The similarity scores served as the basis for generating document recommendations, prioritizing those with the highest similarity values. The system was evaluated in two ways: first, by analyzing the cosine similarity distribution; and second, by assessing recommendation relevance using Precision@k and Recall@k. This approach demonstrated that the fine-tuned classification model could also effectively support content-based recommendation tasks within the academic document domain.

2.1 Data Collection

The dataset used in this study consists of 2000 scientific documents written in English. These documents were manually collected from institutional repositories of various universities in Indonesia. All documents fall within the fields of computer science and informatics. Each document includes metadata such as title, abstract, author, assigned research lab, keywords, year of publication, and source link.

The selection criteria required that the documents be relevant to the field of computer science, written in English, and include complete metadata, especially in the title, abstract, and keyword sections. The lab attribute was not originally present in the metadata and was manually assigned by the researcher. Document labeling was conducted by analyzing the content of each document to determine its primary research focus. Based on this analysis, the documents were grouped into three main categories, namely Embedded Systems with 480 documents, Enterprise Systems with 684 documents, and Intelligent Systems with 836 documents.

To provide an overview of the document distribution across these categories, a graphical summary is presented in Figure 3. This visualization shows how the academic content is distributed among the selected categories.

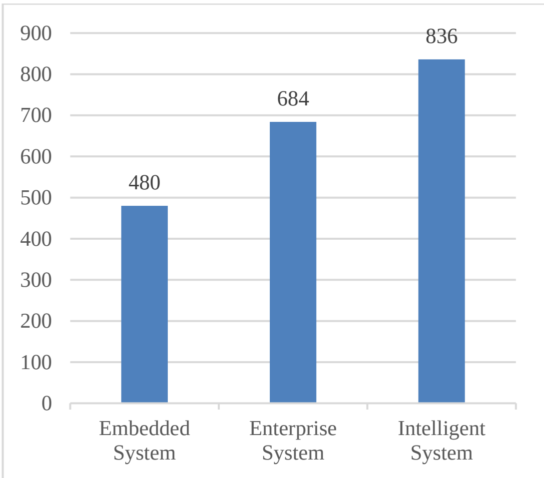


Figure 3. Distribution of Data Entries Based on Labs

As shown in Figure 3, the Intelligent Systems category contains the largest number of documents, followed by Enterprise Systems and Embedded Systems. This distribution reflects natural differences in research focus across the collected documents. The dataset was used in its original form for both training and evaluation. Model performance was assessed separately for each category using standard classification metrics.

The labeling process was conducted through systematic content analysis of the title, abstract, and keywords of each document. The three categories were defined based on commonly recognized research areas in the field of computer science. Documents that discussed hardware, microcontrollers, real-time systems, networking architectures, or cybersecurity frameworks were categorized as Embedded Systems [30]. Documents that focused on enterprise resource planning, organizational information systems, or business process modeling were classified as Enterprise Systems [31]. Documents that addressed artificial intelligence, machine learning, or autonomous systems were labeled as Intelligent Systems [32]. A consistent internal guideline was applied throughout the labeling process to ensure reliability and validity.

The selection of Embedded Systems, Enterprise Systems, and Intelligent Systems as the core categories in this study was based on their prominence as major subfields in computer science. Each category represents a distinct technological focus that is commonly found in academic programs and institutional research structures. These three categories were also the most frequently occurring themes in the collected dataset. They include a wide range of topics such as Internet of Things, networking security, enterprise software, machine learning, and decision support systems. Although computer science includes many other areas, these three categories were considered sufficient to represent the diversity and complexity of topics typically found in academic publications within this field.

2.2 Data Preprocessing

The preprocessing stage began with selecting features to prepare the input for the classification model. Three main attributes were selected from each document, including the title, abstract, and keywords. These elements were combined into a single unified text feature to capture the overall semantic content. The merged text was used as the primary input for the BERT-based model. Classification labels were predefined and converted into numerical form to align with the machine learning process. Each category was encoded using the mapping where Embedded Systems (0), Enterprise Systems (1), and Intelligent Systems (2).

Following the combination of text features and encoding of class labels, the dataset was organized for the tokenization and training process. The goal of this step was to ensure that the input format was consistent and suitable for BERT's architecture. All transformations were applied uniformly to the entire dataset to maintain reproducibility and data integrity. The resulting dataset was then passed to the training pipeline for further processing by the classification model.

2.3 Data Splitting

The data is divided into three subsets, with 70 percent for training, 10 percent for validation, and 20 percent for testing. This allocation is intended to balance the model training and evaluation process, as presented in Table 1.

Table 1. Data Splitting

Label	Train	Validation	Test
Embedded System	336	48	96
Enterprise System	479	68	137
Intelligent System	586	84	167
Total	1400	200	400

2.4 Tokenization and Custom Dataset

After splitting the data, the text is tokenized using the BERT-based-uncased tokenizer from Huggingface Transformers with padding and truncation (max_length = 512).

The tokenized data is converted into a numeric tensor format using the torch.utils.data.Dataset module, which allows for efficient batch processing, shuffling, and ensures stability in the training pipeline.

2.5 Setting Random Seed and Deterministic Configuration

To ensure reproducibility, a random seed is set at the beginning of training. This seed is applied across all random components of the pipeline, including Python libraries, NumPy, and PyTorch.

Deterministic settings are also enabled on the CUDA backend to avoid variations in results due to hardware optimizations.

2.6 Fine-Tuning the BERT Model

The fine-tuning process in this study utilized the pre-trained BERT-base-uncased model provided by Hugging Face through the transformers library. This model consists of 12 transformer encoder layers, 768 hidden units, and 12 attention heads, and has been pre-trained on large English corpora including the BookCorpus and English Wikipedia. It was selected due to its proven effectiveness in various downstream text classification tasks and its broad adoption in academic research [33].

A total of twelve hyperparameter combinations were explored to determine the optimal configuration for the classification task. These combinations varied the learning rate, batch size, and optimizer, while the number of epochs was fixed at four. The full configuration space is presented in Table 2.

Table 2. Fine-Tuning Hyperparameter Settings

Parameter	Value (s)
Epochs	4
Learning Rate	2e-5, 3e-5, 5e-5
Batch Size	16, 32
Optimizer	Adam, AdamW

The selection of hyperparameters was informed by both prior research and exploratory testing. The number of epochs was fixed at four, based on findings that BERT models typically converge efficiently within three to five epochs for classification tasks [23]. Meanwhile, training beyond this range often leads to overfitting or provides only marginal performance improvements that do not justify the additional computational cost [34].

The learning rates of 2e-5, 3e-5, and 5e-5 were selected based on best practices in BERT fine-tuning, where smaller learning rates have consistently produced stable convergence and good generalization. Similarly, batch sizes of 16 and 32 were chosen for their balance between GPU memory efficiency and training stability. These choices are in line with common configurations reported in prior transformer-based classification tasks and have been shown to yield reliable results across a variety of benchmark datasets [35].

Both Adam and AdamW optimizers were used in the experiments. While Adam is a widely accepted default optimizer in deep learning, AdamW has been shown to provide better generalization and weight decay behavior in transformer-based models, including BERT [36].

All configurations were tested during the experimental phase to identify the most effective settings for the classification task. Detailed evaluation results and comparisons are presented in the following section.

2.7 Feature Extraction for Embedding

After obtaining the best model from fine-tuning, feature extraction is performed to generate embedding representations for each document. These embeddings are derived from the last hidden state of the BERT model using the vector from the special [CLS] token, which represents the entire text. The input text consists of the combined title, abstract, and keywords, concatenated into a single string to capture the document's semantic context.

This process is applied to all datasets (training, validation, and testing), with a maximum input length of 512. Tokenization and model invocation are carried out using the saved best fine-tuning results. The resulting embeddings have a fixed dimension of 768 and are used as numerical feature representations for measuring semantic similarity between documents in the recommendation system.

2.8 Recommendation System Based on Cosine Similarity

The 768-dimensional embeddings generated by the BERT model were used to build a document recommendation system based on semantic similarity. Document similarity was calculated using cosine similarity, a method that measures the directional closeness between two embedding vectors in high-dimensional space. This method was chosen due to its high effectiveness in representing semantic relationships without relying on the magnitude of the vectors, making it particularly suitable for comparing contextual embeddings produced by transformer-based models such as BERT.

Unlike other similarity measurement methods that may depend on exact keyword matches or vector magnitude, cosine similarity focuses on the angle between vectors. This allows the method to capture subtle semantic proximity, even when documents differ in length or vocabulary. Its robustness in handling high-dimensional semantic spaces, scalability for large corpora, and effectiveness in identifying contextual similarity make it one of the most widely used approaches in content-based recommendation systems and semantic information retrieval tasks [37].

2.9 Model Evaluation

The model was evaluated based on two components, namely the multi-label classification task and the document recommendation system. During training and validation, the model's performance was assessed using loss values, accuracy, and F1-score across all hyperparameter combinations. After selecting the best-performing model, testing was carried out to evaluate its generalization using accuracy, precision, recall, and F1-score on unseen data [38]. To measure document similarity, cosine similarity was applied to the embedding vectors generated by the BERT model [39]. The relevance of the recommendation results was then evaluated using precision@k and recall@k at various values of k, which are standard metrics in content-based information retrieval and recommendation systems [40].

3. Results and Discussions

The results are based on the training and evaluation of the BERT-based document classification model, as well as the development of a recommendation system using semantic similarity. The evaluation was conducted using performance metrics such as loss, accuracy, F1-score, and the confusion matrix. The discussion begins with an evaluation of the classification model's performance, followed by an analysis of the best-performing model selected from multiple experiments. It then presents the results of the recommendation system, which assesses the relevance of suggested documents based on semantic similarity. This section concludes with a discussion of the main findings and the limitations encountered during the study.

3.1 Model Classification Evaluation

The initial evaluation focused on analyzing the training and validation loss trends across various hyperparameter combinations. This evaluation aimed to assess the model's stability and help identify signs of overfitting or underfitting. The training results based on model performance across different hyperparameter settings are presented in Table 3.

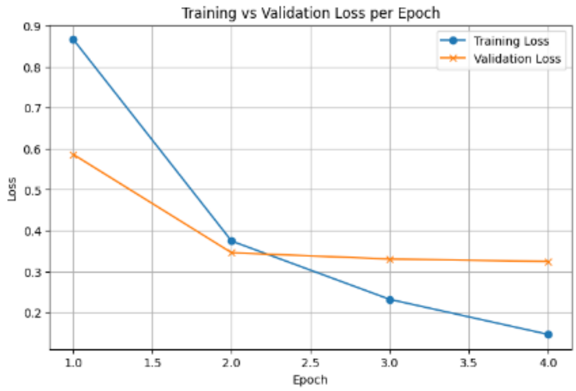
Table 3 Model Performance Across Hyperparameter Variations

Learning Rate, Batch Size, & Optimizer	Training Loss	Validation Loss	Accuracy	F1-Score
2e-5 16 Adam	0.826200	0.485871	0.815000	0.810515
	0.334400	0.321132	0.890000	0.889936
	0.196100	0.393030	0.860000	0.859654
	0.136300	0.350090	0.890000	0.889732
3e-5 16 Adam	0.785900	0.492213	0.810000	0.808518
	0.315600	0.381498	0.885000	0.884611
	0.184000	0.367538	0.895000	0.894964
	0.106800	0.372504	0.890000	0.889137
5e-5 16 Adam	0.748900	0.439147	0.845000	0.844412
	0.350500	0.365691	0.875000	0.875109
	0.202200	0.401930	0.890000	0.889901
	0.111400	0.394944	0.890000	0.889610
2e-5 32 Adam	0.937300	0.604783	0.830000	0.827521
	0.435600	0.381067	0.880000	0.880109
	0.262400	0.349855	0.875000	0.874463
	0.192100	0.346241	0.875000	0.874478
3e-5 32 Adam	0.878300	0.554332	0.805000	0.805035
	0.384100	0.341769	0.890000	0.890342
	0.237000	0.345852	0.880000	0.879755
	0.146800	0.339836	0.885000	0.884488
5e-5 32 Adam	0.835100	0.392832	0.865000	0.863937
	0.329600	0.335676	0.870000	0.870628
	0.205100	0.322586	0.900000	0.899913
	0.109800	0.348229	0.890000	0.889882
2e-5 16 AdamW	0.834800	0.483458	0.810000	0.809746
	0.353500	0.353211	0.900000	0.900030
	0.213300	0.403682	0.855000	0.855051
	0.145200	0.362393	0.870000	0.869575
3e-5 16 AdamW	0.779100	0.515965	0.800000	0.799978
	0.329600	0.345802	0.885000	0.884936
	0.188800	0.424281	0.880000	0.879202
	0.115900	0.361434	0.865000	0.864397
5e-5 16 AdamW	0.715600	0.465913	0.835000	0.834646
	0.332100	0.373791	0.885000	0.884915
	0.169800	0.442836	0.870000	0.869838
	0.096700	0.428430	0.885000	0.884626
2e-5 32 AdamW	0.930100	0.607308	0.825000	0.821753
	0.439500	0.380010	0.880000	0.879806
	0.268900	0.355459	0.870000	0.869656
	0.200100	0.353700	0.880000	0.879344
3e-5 32 AdamW	0.866500	0.585842	0.765000	0.762503
	0.374400	0.345882	0.880000	0.880889
	0.232000	0.330514	0.890000	0.889939
	0.146600	0.324657	0.895000	0.894765
5e-5 32 AdamW	0.826700	0.433204	0.865000	0.863107
	0.326700	0.374099	0.890000	0.889978
	0.194100	0.355610	0.880000	0.879755
	0.102100	0.361864	0.880000	0.879585

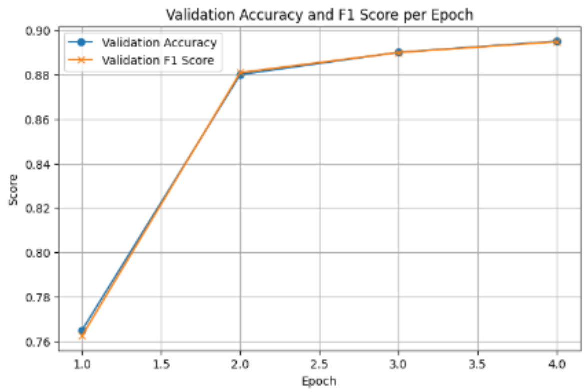
Based on the results in Table 3, it is evident that certain configurations led to imbalanced performance, where the training loss was significantly low while the validation loss remained high. This indicates overfitting, which occurs when the model becomes too tailored to the training data and loses its generalization capability for unseen validation data. For example, the configuration with a learning rate of 2e-5, batch size of 16, and Adam optimizer exhibited such a trend.

However, there were also configurations that demonstrated good fitting, characterized by a balanced relationship between training and validation loss, along with high accuracy and F1-score values. Among all the tested combinations, the configuration with a learning rate of 3e-5, batch size of 32, and the AdamW optimizer showed the best performance. This model not only achieved a low validation loss (0.324657) but also reached an accuracy of 89.5% and an F1-score of 0.894765 among the highest values across all configurations. These results indicate that the model was able to learn effectively without experiencing significant overfitting.

To support these quantitative findings, a visualization of the best-performing model's performance during training was conducted. This visualization illustrates the comparison between training and validation loss, as well as the comparison between validation accuracy and F1-score across each epoch, as shown in Figure 4.



(a)



(b)

Figure 4. Performance of the Best Model During Training. (a) Training and Validation Loss. (b) Validation Accuracy and F1 Score

The visualization in Figure 4 shows that both the training and validation loss exhibit a steadily decreasing trend with a relatively small gap, indicating a well-conducted training process without significant signs of overfitting. Moreover, the validation accuracy and F1-score consistently increased and remained stable at high values towards the end of the training, suggesting that the model is not only accurate but also balanced in handling class distributions within the dataset.

Once the best-performing model was obtained, it was evaluated using the test dataset to assess its generalization capability on previously unseen data. The test results are illustrated by the confusion matrix in Figure 5, which depicts the distribution of predictions compared to the true labels.

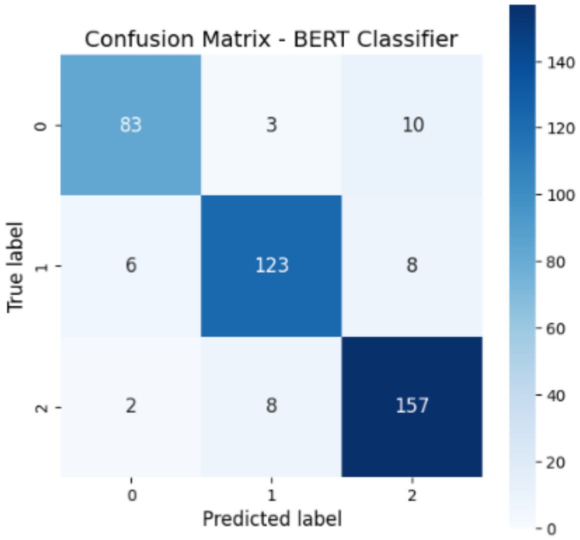


Figure 5. Confusion Matrix of the Best Model

As shown in the confusion matrix in Figure 5, the model demonstrates excellent overall classification performance. Most predictions lie along the main diagonal, indicating that the majority of instances were correctly classified. For example, the Intelligent System class (2) had 157 correct predictions out of 167 samples. Similarly, the Enterprise System class (1) and the Embedded System class (0) achieved 123 and 83 correct predictions, respectively.

To provide a quantitative overview of the model's performance, four primary metrics were used, including accuracy, precision, recall, and F1-score. The evaluation results on the test dataset are presented in Table 4.

Table 4 Evaluation Results of the Best Model on the Test Dataset

Metric	Value (%)
Accuracy	91
Precision	91
Recall	90
F1-Score	90

Based on the evaluation results shown in Table 4, the best-performing model achieved an accuracy of 91%, with precision and recall of 91% and 90%, respectively, and an F1-score of 90%. These results indicate that the model demonstrates high and balanced classification performance across various document categories.

Nevertheless, further evaluation using the confusion matrix revealed that not all categories were perfectly distinguishable by the model. One of the main challenges identified was in differentiating between the Embedded Systems and Intelligent Systems categories. According to the prediction distribution, the model tended to struggle in distinguishing these two categories. This is likely due to overlapping terminology and subject matter commonly found in documents from both categories. Generally, Embedded Systems involve topics such as hardware, microcontrollers, networking, and real-time systems, while Intelligent Systems focus more on artificial intelligence, machine learning, and autonomous systems.

In real-world applications, many documents are interdisciplinary in nature, incorporating elements from both categories. For example, an AI-based control system implemented on embedded hardware contains characteristics of both Embedded Systems and Intelligent Systems. When such documents are processed by the BERT model, the resulting embeddings tend to share semantic similarities and are positioned closely in the vector space. This poses a challenge for classification, as the main context of a document is not always explicitly stated, making it difficult for the model to assign the correct label.

Despite these challenges in differentiating semantically overlapping categories, evaluation results from both the validation and testing phases indicate that the fine-tuned BERT model with a learning rate of 3e-5, batch size of 32, and AdamW optimizer was the best-performing model in this study. This model was selected as the foundation for the subsequent recommendation system development phase due to its consistently high and stable performance across all evaluation metrics.

3.2 Recommendation System Evaluation

The recommendation system in this study adopts a content-based approach by leveraging the embedding representations from a fine-tuned BERT model (bert-base-uncased). Each academic document is encoded into a 768-dimensional vector by integrating information from the title, abstract, and keywords, providing a comprehensive representation of the document's semantic meaning. The vector representations of several sample documents are illustrated in Figure 6.

As shown in Figure 6, each document exhibits a unique distribution pattern across specific dimensions, reflecting the semantic differences among documents. This representation enables the system to match documents based on contextual similarity rather than relying solely on word-level or surface-level similarities.

The next step in the recommendation process is to compute the semantic similarity between documents using the cosine similarity method, where values approaching 1 indicate a high degree of semantic similarity. The results of these similarity computations are visualized in the form of a matrix in Figure 7, which illustrates the relationships between document pairs

text	Dimensi 1	Dimensi 2	Dimensi 3	Dimensi 4	Dimensi 5	...	Dimensi 764	Dimensi 765	Dimensi 766	Dimensi 767	Dimensi 768
doc_1	0.032170	0.024857	1.191476	0.500296	0.185689	...	1.111483	-0.021955	0.250330	-1.022530	0.418127
doc_2	-1.312643	0.618225	0.009292	0.431450	0.491653	...	-0.045226	-0.528527	0.267358	-1.321002	0.648892
doc_3	-0.725290	-0.163856	1.174989	0.674043	-0.147738	...	1.278575	-0.024541	0.331284	-1.341356	0.632876
doc_4	0.244675	-0.267485	1.232532	0.744465	-0.310778	...	1.012758	0.183469	0.780050	-0.847422	0.357336
doc_5	0.594113	-0.248594	1.129304	0.438653	-0.027481	...	1.382339	-0.005897	0.892328	-0.742749	0.454142
...	...	...	...	...	...	...	...	...	...	...	...
doc_1996	-1.544663	0.400078	-1.539793	-0.557561	-0.107372	...	-0.470894	-0.344774	0.644529	-0.401080	-0.080311
doc_1997	-1.251546	0.334643	-1.198278	-0.664890	-0.119351	...	-0.593014	-0.031748	0.856011	-0.854143	-0.102259
doc_1998	-1.243502	0.476591	-1.235348	-0.662608	0.026938	...	-0.516612	-0.203609	0.781956	-0.562444	-0.351607
doc_1999	0.999321	0.852400	0.137000	0.116835	0.184080	...	0.814337	-0.469843	0.064577	0.385914	-0.511784
doc_2000	-0.013679	-0.163064	0.833916	0.875799	-0.015701	...	0.877798	0.048890	0.575906	-0.709355	0.546688

Figure 6. 768-Dimensional Embedding Vector Representation for Academic Documents

	text	doc_1	doc_2	doc_3	doc_4	doc_5	...	doc_1996	doc_1997	doc_1998	doc_1999	doc_2000
0	doc_1	1.000000	0.440044	0.863967	0.893431	0.873979	...	-0.146507	-0.109288	-0.135805	0.342897	0.936843
1	doc_2	0.440044	1.000000	0.482499	0.305603	0.322830	...	0.266706	0.280328	0.228789	0.196452	0.390887
2	doc_3	0.863967	0.482499	1.000000	0.864736	0.803441	...	-0.005542	0.058898	0.020320	0.270138	0.849647
3	doc_4	0.893431	0.305603	0.864736	1.000000	0.929965	...	-0.202329	-0.157232	-0.181987	0.369776	0.928788
4	doc_5	0.873979	0.322830	0.803441	0.929965	1.000000	...	-0.251983	-0.213245	-0.242528	0.453791	0.898010
...	...	...	...	...	...	...	...	...	...	...	...	...
1995	doc_1996	-0.146507	0.266706	-0.005542	-0.202329	-0.251983	...	1.000000	0.930862	0.929958	-0.273040	-0.135703
1996	doc_1997	-0.109288	0.280328	0.058898	-0.157232	-0.213245	...	0.930862	1.000000	0.961891	-0.311310	-0.103287
1997	doc_1998	-0.135805	0.228789	0.020320	-0.181987	-0.242528	...	0.929958	0.961891	1.000000	-0.297965	-0.134855
1998	doc_1999	0.342897	0.196452	0.270138	0.369776	0.453791	...	-0.273040	-0.311310	-0.297965	1.000000	0.307069
1999	doc_2000	0.936843	0.390887	0.849647	0.928788	0.898010	...	-0.135703	-0.103287	-0.134855	0.307069	1.000000

Figure 7. Cosine Similarity Matrix Between Documents

As shown in Figure 7, documents such as doc_1 exhibit a cosine similarity score of 0.893 with doc_4 and 0.874 with doc_5, indicating strong topical alignment despite differences in textual expression. Similarly, doc_2000 demonstrates high semantic proximity with doc_1994 and doc_1995, each with similarity scores exceeding 0.93. These results indicate that the system is capable of identifying deep thematic relationships between documents based on meaning rather than relying solely on explicit keyword matching.

To evaluate the quality of the recommendation system, Precision@k and Recall@k metrics were used. The evaluation compared two approaches, namely a fine-tuned BERT model and the original BERT base model without additional tuning. The results are presented in Table 5.

The results in Table 5 demonstrate that the fine-tuned model consistently achieved Precision@k scores above 92% for k values ranging from 5 to 30, significantly outperforming the base BERT model. This suggests that the top-ranked recommended documents are highly relevant to the source document. Although Recall@k values were initially low (due to only a small fraction of documents being considered), they increased with larger k values, reaching 1.0 when all documents were evaluated. This indicates that the system is capable of retrieving all relevant documents at a larger scale.

Table 5. Precision@k and Recall@k Comparison Between Fine-Tuned and Base BERT Models

Top-k	Precision@k (fine tuned)	Recall@k (fine tuned)	Precision@k (base)	Recall@k (base)
5	0.9313	0.0070	0.7066	0.0052
10	0.9290	0.0139	0.6829	0.0101
15	0.9295	0.0209	0.6670	0.0148
20	0.9298	0.0279	0.6558	0.0194
30	0.9298	0.0418	0.6392	0.0283
1999	0.3490	1.0000	0.3490	1.0000

3.3 Comparison with Related Studies

This study is compared with prior research that employed BERT-based approaches for document classification and recommendation. One study reported that a fine-tuned BERT model achieved an F1-score of 93.25% in classifying policy documents from various institutions; however, the study did not involve the development of a recommendation system [23]. Another study applied BERT to classify extremist and hoax-related content, achieving an accuracy of 99.71%. However, the dataset used in that study consisted of social media texts, which are generally short and structurally simple, in contrast to the long and complex academic documents used in this research [24].

Another study developed a scientific document recommendation system using BERT embeddings derived from author-provided keywords. The system showed higher relevance compared to index-based search engines such as Scopus. Nevertheless, this approach did not include a classification component and did not evaluate the effectiveness of document grouping [29]. These studies indicate that existing approaches tend to focus on either classification or recommendation, without integrating both functionalities into a unified framework.

In contrast, the system developed in this study integrates document classification and recommendation within a single fine-tuned BERT-based architecture. The fine-tuning process involved evaluating various hyperparameter combinations, with the optimal configuration found at a learning rate of 3e-5, batch size of 32, and AdamW optimizer. During training, the model achieved an accuracy of 89.5% and an F1-score of 0.894765. On the test set, it reached an accuracy of 91% and an F1-score of 90%. Furthermore, the recommendation system consistently produced high performance, with Precision@k values exceeding 92% for k ranging from 5 to 30. These findings demonstrate the system's ability to deliver both competitive classification accuracy and semantically relevant recommendations for complex and interdisciplinary academic documents.

4. Conclusions

The conclusion of this study highlights the successful development of a scientific document classification and recommendation system based on a fine-tuned BERT model. The system was designed to categorize academic documents in the field of computer science into three main classes, namely Embedded Systems, Enterprise Systems, and Intelligent Systems. The optimal model configuration was achieved with a learning rate of 3e-5, batch size of 32, and the AdamW optimizer. The model demonstrated high and balanced classification performance, with test results showing 91% accuracy, 91% precision, 90% recall, and 90% F1-score. During training, the model maintained stability and did not exhibit overfitting, as reflected in the consistently decreasing trends in training and validation loss with minimal gap. The training accuracy of 89.5% and F1-score of 0.894765 closely aligned with test results, indicating strong generalization. The recommendation system utilized BERT-based semantic embeddings and cosine similarity to measure document relevance. It consistently achieved Precision@k values above 92% for k ranging from 5 to 30, significantly outperforming the base BERT model. Although Recall@k started low due to limited top-k evaluation, it increased with larger k values and reached 1.0 when all documents were considered, demonstrating the system's broad coverage. Misclassifications primarily occurred at the semantic boundaries between categories, particularly between Embedded Systems and Intelligent Systems. To improve classification accuracy, the integration of contextual attributes such as laboratory affiliation or application domain is recommended. Additionally, exploring alternative embedding models such as RoBERTa, DistilBERT, or Sentence-BERT may offer promising enhancements for future system development.

References

[1] A. S. Rahmatullah, E. Mulyasa, S. Syahrani, F. Pongpalilu, and R. E. Putri, "Digital Era 4.0: The Contribution to Education and Student Psychology," *Linguistics and Culture Review*, vol. 6, no. 3, pp. 89–107, 2022, doi: 10.21744/lingcure.v6ns3.2064.

[2] R. I. Mukhamediev *et al.*, "Review of Artificial Intelligence and Machine Learning Technologies: Classification, Restrictions, Opportunities and Challenges," *Mathematics*, vol. 10, no. 2552, pp. 1–25, 2022, doi: 10.3390/math10152552.

[3] M. Musaev, M. Rakhmatullaev, S. Normatov, K. Shukurov, and M. Abdullaeva, "Integrated Intelligent System for Scientific and Educational Information Retrieval," in *Proceedings of the 15th International Scientific and Practical Conference*, Rezekne Academy of Technologies, 2024, pp. 212–219. doi: 10.17770/etr2024vol2.8028.

[4] E. F. Okagbue *et al.*, "A comprehensive overview of artificial intelligence and machine learning in education pedagogy: 21 Years (2000–2021) of research indexed in the scopus database," *Social Sciences and Humanities Open*, vol. 8, 2023, doi: 10.1016/j.ssaho.2023.100655.

[5] K. A. Hambarde and H. Proenca, "Information Retrieval: Recent Advances and Beyond," *IEEE Access*, vol. 11, pp. 76581–76604, 2023, doi: 10.1109/ACCESS.2023.3295776.

[6] M. Zhang, J. Fan, A. Sharma, and A. Kukkar, "Data mining applications in university information management system development," *Journal of Intelligent Systems*, vol. 31, pp. 207–220, 2022, doi: 10.1515/jisys-2022-0006.

[7] C. K. Kreutz and R. Schenkel, "Scientific paper recommendation systems: a literature review of recent publications," *International Journal on Digital Libraries*, vol. 23, pp. 335–369, 2022, doi: 10.1007/s00799-022-00339-w.

[8] H. Ezaldeen, S. K. Bisoy, R. Misra, and R. Alatrash, "Semantics aware intelligent framework for content-based e-learning recommendation," *Natural Language Processing Journal*, vol. 3, pp. 1–16, 2023, doi: 10.1016/j.nlp.2023.100008.

[9] S. Jha, C. V. Mahamuni, and I. K. Garewal, "Natural Language Processing: A Literature Survey of Approaches, Applications, Current Trends, and Future Directions," in *2024 Asian Conference on Intelligent Technologies, ACOIT 2024*, Institute of Electrical and Electronics Engineers Inc., 2024. doi: 10.1109/ACOIT62457.2024.10941573.

[10] N. M. Gardazi, A. Daud, M. K. Malik, A. Bukhari, T. Alsahfi, and B. Alshemaimri, "BERT applications in natural language processing: a review," *Artif Intell Rev*, vol. 58, no. 166, pp. 1–49, 2025, doi: 10.1007/s10462-025-11162-5.

[11] A. Rahali and M. A. Akhloufi, "End-to-End Transformer-Based Models in Textual-Based NLP," *AI*, vol. 4, pp. 54–110, 2023, doi: 10.3390/ai4010004.

[12] E. Kotei and R. Thirunavukarasu, "ASystematic Review of Transformer-Based Pre-Trained Language Models through Self-Supervised Learning," *Information*, vol. 14, pp. 1–25, 2023, doi: 10.3390/info14030187.

[13] H. Wang, J. Li, H. Wu, E. Hovy, and Y. Sun, "Pre-Trained Language Models and Their Applications," *Engineering*, vol. 25, pp. 51–65, 2023, doi: 10.1016/j.eng.2022.04.024.

[14] W. Zhang, Y. Yang, T. Akilan, Q. M. J. Wu, and T. Liu, "Fast Transfer Learning Method Using Random Layer Freezing and Feature Refinement Strategy," *IEEE Trans Cybern*, vol. 55, no. 1, pp. 234–246, 2024, doi: 10.1109/TCYB.2024.3483068.

[15] L. Wang *et al.*, "Parameter-efficient fine-tuning in large language models: a survey of methodologies," *Artif Intell Rev*, vol. 58, pp. 1–64, 2025, doi: 10.1007/s10462-025-11236-4.
[16] A. H. Salman and W. Al-Jawher, "A New Multi-class Classification Method Based on Machine Learning to Document Classification," in *16th International Conference on Developments in eSystems Engineering (DeSE)*, IEEE Xplore, 2023, pp. 605–610. doi: 10.1109/DeSE60595.2023.10468839.
[17] R. K. Shah, S. Kumar, and Shashank, "Multilabel News Category Classification using Machine Learning," in *Proceedings of the 8th International Conference on Communication and Electronics Systems (ICCES 2023)*, IEEE Xplore, 2023, pp. 1245–1250. doi: 10.1109/ICCES57224.2023.10192826.
[18] M. Memoon, M. Umar, S. Rani, M. Khalid, and S. Shaheen, "Twitter News Classification Using Machine Learning," in *2024 International Conference on Engineering & Computing Technologies (ICECT)*, IEEE Xplore, 2024, pp. 1–7. doi: 10.1109/ICECT61618.2024.10581173.
[19] A. Onan and H. Alhumyani, "Contextual Hypergraph Networks for Enhanced Extractive Summarization: Introducing Multi-Element Contextual Hypergraph Extractive Summarizer (MCHES)," *Applied Sciences*, vol. 14, pp. 1–25, 2024, doi: 10.3390/app14114671.
[20] A. R. I. Fauzy and E. B. Setiawan, "Detecting Fake News on Social Media Combined with the CNN Methods," *JURNAL RESTI (RekayasaSistemdanTeknologiInformasi)*, vol. 7, no. 2, pp. 271–277, 2023, doi: 10.29207/resti.v7i1.4889.
[21] S. A. Zikrina and Fitriyani, "Advancing Hate Speech Detection in Indonesian Language Using Graph Neural Networks and TF-IDF," *Jurnal RESTI*, vol. 9, no. 1, pp. 137–145, 2025, doi: 10.29207/resti.v9i1.6179.
[22] A. A. Nabil, D. Das, M. S. Salim, S. Arifeen, and H. M. A. Fattah, "Bangla Emergency Post Classification on Social Media using Transformer Based BERT Models," in *6th International Conference on Electrical Information and Communication Technology (EICT)*, IEEE Xplore, 2023, pp. 1–6. doi: 10.1109/EICT61409.2023.10427900.
[23] B. Yu, C. Deng, and L. Bu, "Policy Text Classification Algorithm Based on Bert," in *11th International Conference of Information and Communication Technology (ICTech)*, IEEE Xplore, 2022, pp. 488–491. doi: 10.1109/ICTech55460.2022.00103.
[24] R. Qasim, W. H. Bangyal, M. A. Alqarni, and A. A. Almazroi, "A Fine-Tuned BERT-Based Transfer Learning Approach for Text Classification," *J Healthc Eng*, vol. 2022, pp. 1–17, 2022, doi: 10.1155/2022/3498123.
[25] M. Bilal and A. A. Almazroi, "Effectiveness of Fine-tuned BERT Model in Classification of Helpful and Unhelpful Online Customer Reviews," *Electronic Commerce Research*, vol. 23, pp. 2737–2757, 2022, doi: 10.1007/s10660-022-09560-w.
[26] M. Chiny, M. Chihab, O. Bencharef, and Y. Chihab, "Netflix Recommendation System based on TF-IDF and Cosine Similarity Algorithms," in *2nd International Conference on Big Data, Modelling and Machine Learning (BML 2021)*, SCITEPRESS, 2022, pp. 15–20. doi: 10.5220/0010727500003101.
[27] S. Al-Otaibi *et al.*, "Cosine similarity-based algorithm for social networking recommendation," *International Journal of Electrical and Computer Engineering (IJECE)*, vol. 12, no. 2, pp. 1881–1892, 2022, doi: 10.11591/ijece.v12i2.pp1881-1892.
[28] C. G. Reswara, J. Nicolas, M. Ananta, and F. I. Kurniadi, "Anime Recommendation System Using Bert and Cosine Similarity," in *4th International Conference on Artificial Intelligence and Data Sciences: Discovering Technological Advancement in Artificial Intelligence and Data Science (AiDAS)*, IEEE Xplore, 2023, pp. 109–113. doi: 10.1109/AiDAS60501.2023.10284693.
[29] S. Sankar, H. Muslihuddeen, S. Ostwal, P. Sathvika, and A. K. Madasamy, "CISER: Customized Institute Specific Search Engine for Retrieving Research Papers," in *4th International Conference on Innovative Trends in Information Technology (ICITIIT)*, IEEE Xplore, 2023, pp. 1–5. doi: 10.1109/ICITIIT57246.2023.10068620.
[30] W. I Alhababi and A. W Mulhim, "What are the Embedded Systems," *Int J Innov Sci Res Technol*, vol. 9, no. 9, pp. 1585–1587, 2024, doi: 10.38124/ijisrt/ijisrt24sep953.
[31] N. Mayasari, E. Andriani, and A. R. Hibrida, "Revolutionizing Business Operations: A Bibliometric Analysis of Enterprise Systems and Organizational Efficiency," *The Eastasouth Journal of Information System and Computer Science*, vol. 1, no. 1, pp. 45–54, 2023, doi: 10.58812/esiscs.v1i01.134.
[32] P. K. Hristova, M. Ivanovic, and B. Diri, "Special issue: Innovations in Intelligent Systems and Applications," *Journal of Information and Telecommunication*, vol. 6, no. 1, pp. 1–5, 2022, doi: 10.1080/24751839.2021.1989242.
[33] M. P. Geetha and D. K. Renuka, "Improving the performance of aspect based sentiment analysis using fine-tuned Bert Base Uncased model," *International Journal of Intelligent Networks*, vol. 2, pp. 64–69, 2021, doi: 10.1016/j.ijin.2021.06.005.
[34] K. N. Prasanthi, R. E. Madhavi, D. N. S. Sabarinadh, and B. Sravani, "A Novel Approach for Sentiment Analysis on social media using BERT & ROBERTA Transformer-Based Models," in *International Conference for Convergence in Technology (I2CT)*, IEEE Xplore, 2023, pp. 1–6. doi: 10.1109/I2CT57861.2023.10126206.
[35] X. Li, X. Wang, and H. Liu, "Research on fine-tuning strategy of sentiment analysis model based on BERT," in *2024 10th International Conference on Smart Computing and Communication (ICSCC)*, IEEE Xplore, May 2021, pp. 798–802. doi: 10.1109/CISCE52179.2021.9445882.
[36] P. Nabila and E. B. Setiawan, "Adam and AdamW Optimization Algorithm Application on BERT Model for Hate Speech Detection on Twitter," in *2024 International Conference on Data Science and Its Applications (ICoDSA)*, IEEE Xplore, 2024, pp. 346–351. doi: 10.1109/ICoDSA62899.2024.10651619.
[37] R. Shaharao *et al.*, "Semantic Similarity in Multi-Source Information Retrieval for Improving Learner Performance," in *International Conference on Smart Computing and Communication (ICSCC)*, IEEE Xplore, 2024, pp. 1–6. doi: 10.1109/ICSCC62041.2024.10690443.
[38] M. Heydarian, T. E. Doyle, and R. Samavi, "MLCM: Multi-Label Confusion Matrix," *IEEE Access*, vol. 10, pp. 19083–19095, 2022, doi: 10.1109/ACCESS.2022.3151048.
[39] B. D. Wijanarko *et al.*, "AI-Based Feature Extraction and Cosine Similarity for Automation of Student Learning Assessment," in *24th International Arab Conference on Information Technology (ACIT)*, IEEE Xplore, 2023. doi: 10.1109/ACIT58888.2023.10453927.
[40] M. S. Sungkar, R. Wulandari, S. Syamsidar, W. Firdaus, A. Andiyan, and R. Rahim, "Enhancing E-commerce Recommendation Accuracy Using KNN and Hybrid Approaches: An Empirical Study," in *Advances in Intelligent Manufacturing and Robotics*, Springer, 2023, pp. 293–300. doi: 10.1007/978-981-99-8498-5_22.